Open Systems Handbook
Caching, Backpressure and Operations

Edition 2.1

A fictional technical handbook written for the local-docs-rag-assistant portfolio demo. It contains no real company, customer or personal data.

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2. Caching

The default caching pattern is cache-aside: the application reads the cache, loads from the source on a miss, then writes the value back.

Write-through is reserved for reference data that changes less than once per hour. Do not use write-through for user-generated content.

Every TTL receives 10% jitter. Without jitter, many keys expire at the same moment and create a load spike.

Cache stampedes are prevented with single-flight: one request recomputes a missing value while the others wait up to 2 seconds. If the wait expires, the caller reads from the source directly.

Negative results may be cached for at most 30 seconds. Longer negative caching makes recovery from a transient failure invisible to users.

Never cache 5xx responses.

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3. Backpressure

Every worker owns a bounded queue with a capacity of 1,000 items.
Unbounded queues hide overload and turn it into latency and memory growth.

When the queue reaches 80% capacity, the worker rejects new work with HTTP 429.
The rejection is intentional load shedding, not an error to retry immediately.

Admission is also controlled by a token bucket that refills 100 tokens per second and holds at most 200 tokens. A request without a token is rejected before it reaches the queue.

When the system is saturated, shed work in this order:

1. Health checks are never shed.
2. Reads are shed before writes.
3. Background jobs are paused first.

Record every rejection with a reason code. A rejected request is a successful protection of the system, not a lost metric.

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4. Observability

Monitor four golden signals: latency, traffic, errors and saturation.
Each dashboard shows p50, p95 and p99 latency; an average hides the experience of the slowest users.

Trace sampling is 10% in normal operation. All error traces are retained.
Metrics are kept for 30 days, traces for 7 days and logs for 21 days.

Alerts must describe user impact and link to a runbook. A CPU alert without a user-facing symptom is a capacity signal, not an incident.

Review dashboards after every incident. Remove panels nobody uses; a dashboard that is too large to read during an incident has negative value.

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5. Capacity planning

Capacity planning uses Little's Law: L = lambda x W, where L is the number of items in the system, lambda is the arrival rate and W is the time each item spends in the system.

Plan for 30% headroom above the forecast peak. Headroom absorbs traffic spikes, deployment overlap and the loss of one availability zone.

Before launch, run a load test at 2x the expected peak. The test must include the slowest dependency, not a mocked fast path.

Review capacity monthly and after any change that increases payload size or fan-out. A forecast older than one quarter is not a forecast; it is a guess.

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6. Change management

Changes affecting more than 5% of traffic require a feature flag. The flag must have an owner and an expiry date; permanent flags become hidden configuration.

The rollback target is 5 minutes from the decision to roll back. Rehearse the rollback before the release, not during the incident.

The standard maintenance window is Tuesday, 03:00-05:00 UTC. Work outside the window requires approval from the incident commander on call.

Emergency changes require incident commander approval and a follow-up review. During a Sev1, the change freeze applies to everything except the approved fix.

Every change is recorded with its version, flag state and rollback decision.

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